Anatolii Kurotych

Software Engineer

9 years of experience building production services for telecom and blockchain networks in both startups and established companies. Core expertise: system-level programming (Rust, C/C++), backend architecture, end-to-end system development on GNU/Linux, technical leadership, and team management. Active open-source contributor.

+380958795130	**akurotych@gmail.com**	**Lutsk, Ukraine**
Linkedin	GitHub	kurotych.com

Technical Skills

Programming Languages:

Primary: Rust, Python, C, C++, Bash

Familiar: Go, TypeScript, JavaScript, Solidity, Lua.

Infrastructure & Cloud: Linux, Docker, AWS, GitHub Actions, CI/CD Pipelines

Databases: PostgreSQL, MySQL, SQLite, Apache Iceberg, Trino

APIs & Interfaces: GraphQL, gRPC, REST API

Systems & Design: Distributed Systems, System Architecture, Concurrency

Work experience

Senior Software Engineer at Helium (Nova Labs)

(04/23 — 04/26) 3 yrs

Industry: Telecommunications, Blockchain

Tech stack: Rust, Python, Bash, PostgreSQL, GraphQL, gRPC, Protobuf, Docker, AWS, GitHub Actions

- Built a radio signal propagation service using **Rust**, **GraphQL**, and **gRPC**, with geospatial processing to support mobile coverage modeling and API-based data access in the Helium network.
- Designed high-throughput ETL pipelines processing hundreds of gigabytes of data daily. Ingested **Protobuf**-encoded data from **AWS S3** into **PostgreSQL** and **Apache Iceberg**, powering systems including the Helium Mobile Explorer and providing access via a **GraphQL API**.
- Drove architecture decisions for distributed Rust services supporting mobile coverage modeling and large-scale telemetry data processing in the Helium network.
- Contributed to the Helium Mobile Oracles (**Rust**) — implemented gRPC services, extended Apache Iceberg integration, and improved system reliability.
- Developed a suite of **Rust**-based services that run autonomously, consume minimal resources, and have operated in production with no incidents.

Tech Lead / Software Engineer at Minterest

(03/21 — 04/23) 2 yrs 1 mo

Industry: Finance (DeFi, Blockchain)

Tech stack: Rust, Substrate (Polkadot SDK), Node.js, Solidity, TypeScript, JavaScript, Bash, AWS, Docker

Tech Lead *(10/21 — 04/23) 1 yr 6 mos*

- Led the development of a DeFi lending platform from scratch to private launch
- Participated in strategic tech discussions
- Provided blockchain (and web) architecture solutions. Authored high-level design (HLD) documents, requirements, and diagrams.
- Collaborated with the team to propose and define the tech stack.
- Maintained project documentation using Confluence.
- Implemented CI/CD. Wrote CI jobs with **CircleCI**, **AWS** infrastructure via **Terragrunt**.
- Reviewed and wrote code in **Node.js, TypeScript, Solidity, JavaScript, and Bash.**
- Led sprint planning, backlog grooming, and engineering task management.

Software Engineer *(03/21 — 10/21) 7 mos*

Achievements/Tasks

- Developed polkadot parachain with **Rust and Substrate**
- Wrote and reviewed Rust code

Software Engineer C/C++/Python at PortaOne

(06/19 — 02/21) 1 yr 9 mos

Industry: Telecommunications

Tech stack: C++, C, Python, CMake

Achievements/Tasks

- Optimized and maintained high-load SIP servers (C++) on **Linux**.
- Modernized codebase using C++17 standards
- Reduced service compilation time significantly.
- Developed framework to test services with **Python**.
- Improved services documentation.

Software Engineer C/C++, Team lead at Demlabs

(07/18 — 04/19) 10 mos

Industry: IT Infrastructure & Networking

Tech stack: C++, Qt, C, Python, CMake

Achievements/Tasks

- Lead Developer and Team Lead of the DiveVPN service (KelvinVPN now).
- Designed system architecture for cross-platform VPN services.

- Implemented a robust finite-state machine (FSM) improving system stability
- Wrote code for high-load sections.
- Packaged and deployed beta releases across Linux, Windows, macOS, and Android
- Established project testing strategy and integrated CI solutions
- Led and managed a team of ~5 developers.
- Contributed to cellframe-sdk.

Software Developer Go/NodeJS at EssentiaOne

(10/17 — 07/18) 10 mos

Industry: Blockchain

Tech stack: Go, NodeJS, Solidity

Achievements/Tasks

- Developed architecture design and wrote code for REST API web service to interact with different blockchains (Bitcoin, Ethereum, etc) and decentralized storages like Swarm(Ethereum)/IPFS/Storj via a unified API based on **Go**.
- Developed a web service to interact with decentralized applications by REST API with **NodeJS**.

Software Developer C/C++ at Demlabs

(01/17 — 10/17) 10 mos

Industry: Industrial Software, IT Infrastructure & Networking

Tech stack: C++, Qt, C, Python, CMake

Achievements/Tasks

- Developed human-machine interface with **C++/Qt**.
- Resolved bugs and optimized performance of Linux VPN server modules written in **C**.
- Enhanced and extended the cross-platform VPN client with new features, performance improvements, and bug fixes using **C++/Qt**.

Education

Lesya Ukrainka Volyn National University (2024-2025) Master's degree in Computer Science

Lutsk National Technical University (2019-2022) Bachelor's degree in Software Engineering

STEP Computer Academy (2016-2017) Software development course (1 year)

Technical College LNTU (2011-2015) Associate Degree in Automotive Maintenance and Engine Repair

Achievements

- Developed the first open-source implementation of Schnorr digital signatures not based on OpenSSL, integrated into Trezor Wallet and many other projects.

- Authored and presented the article *“Optimizing the Process of ER Diagram Creation with PlantUML”*, published in the *Proceedings of the 7th Workshop for Young Scientists in Computer Science & Software Engineering (CS&SE@SW 2024).* Read paper

Personal projects

- sqlant (2022) - CLI tool that generates textual PlantUML ER diagrams from live SQL connection strings to assist with database documentation and visualization.
- sipmsg (2019) - Parse and validate SIP Messages according to RFC3261 with Rust.

Certificates and Recommendations

Can be seen on my Linkedin page

Languages

Ukrainian(native), English(B2 – upper-intermediate)